Stronger dissipation, longer memory: engineering the jump operators of quantum reservoirs

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Dissipation is usually treated as the clock of a quantum reservoir computer: the more the reservoir leaks into its environment, the faster it forgets. We show that this depends on how it leaks. In the spin reservoirs we study, the same total coupling to the environment has opposite effects. If each qubit leaks on its own, stronger leaking erases memory faster. If the qubits leak through one shared channel, stronger leaking makes memory last longer. The reason is interference: the decay paths of the qubits cancel for some states, which cannot leak at all, and strong leaking holds the reservoir inside them. A long memory is a trap of its own, because every new input nudges what is stored and old inputs blend with the ones that follow. We prove that in slowly forgetting reservoirs, on average, the number of inputs blended together is how hard one input nudges the stored state times how long the memory lasts. Writing each input more gently as the memory grows, half as strongly for a memory four times longer, keeps single past inputs readable over a range of delays that grows with the leaking: we prove this in a three-spin reservoir, where coherence between dark states carries the memory. The memory becomes longer but thinner. How a quantum system leaks decides not only how fast it forgets, but what it keeps and in what form.

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1 Introduction

A reservoir computer processes a stream of inputs with a fixed dynamical system and trains only a linear readout [1, 2]. In quantum reservoir computing (QRC) the system is quantum [3], and how it keeps and loses information about past inputs decides what it can compute: recent inputs must stay available, while older inputs and the initial state must fade. Unitary dynamics does not forget, so dissipation is part of the design [4]. Damping, absorption, measurement and resets all create or regulate this fading memory [5, 6, 7, 8, 9, 10, 11, 12, 13, 14]. Tuning the rate of independent qubit relaxation improves a driven spin reservoir [6], and related work tunes the strengths of the drive, the interactions and the damping [15, 16]. In these studies the environment acts as a clock: the more the reservoir leaks, the faster it forgets.

An environment specifies more than a rate. Its jump operators decide how the reservoir leaks: each qubit into its own channel, all qubits through one shared channel, pairs together, or while excitations hop from qubit to qubit. Choosing them is the craft of dissipation engineering, which prepares and protects quantum states and even computes with them [17, 18, 19, 20]. A shared channel makes the decay paths of the qubits interfere: some collective states decay faster, while others are dark and cannot decay at all, as in subradiance [21, 22, 23]. Earlier QRC studies tuned the rates of local channels [6], compared local noise types in gate-based reservoirs [24, 25], found that stronger photon loss favours redundant encoding in coupled Kerr oscillators [26], or read a static classifier out of a decoherence-free subspace [27]. How the way a reservoir leaks shapes its memory of a time series has not been compared as the

loss strength is varied at equal coupling to the environment.

Here we show that it decides the outcome. At the same total coupling, separate channels make stronger leaking erase memory faster, while one shared channel makes stronger leaking preserve it longer (Fig. 1). The reason is interference. The shared channel cannot reach its dark states, and strong leaking acts like constant checking that holds the reservoir inside them, the quantum Zeno effect [28, 29, 30]; what still escapes falls as $1/\gamma$. A model of the dark states alone, with nothing fitted, reproduces the memory of the full reservoir, and erasing them destroys it.

Lasting longer is not the same as staying readable. Every new input nudges the stored state, so a long-lived record is blended with everything that came after it (Fig. 1d). We prove a simple law: on average, the number of inputs blended together, the order of the memory, is the mixing fraction of one input, how hard it nudges the stored state, times the memory time. The fix follows from the law. Write more gently as the memory grows, with an input amplitude that falls as one over the square root of the memory time, and let the input reach the dark states directly. Single past inputs then stay readable over a range of delays, the number of past steps that hold 90% of the recall, that grows in proportion to the loss rate; in a three-spin reservoir we prove this and show that coherence between dark states carries the memory. The price is that the memory becomes longer but thinner: each delay is recalled more faintly, the total cannot exceed what the readout holds, and ordinary local noise limits how far the gain goes.

Memory here means what a trained linear readout recovers from exact expectation values. The mixing law holds for a class of reservoirs; the spin results hold for the reservoirs we study, with specified Hamiltonians and jump operators. The first version of this study [31] compared the same jump families on four benchmark tasks (Appendix D) and local with collective relaxation from four to eight spins; the dark-state mechanism, the mixing law, the writing rule and the proofs are new here.

2 Results

2.1 Same total leaking, different channels

We use the driven spin reservoir of Ref. [6]. In step t an input $u_t \in [0, 1]$, drawn independently, is held for a time $\Delta t = 1/2$ while $N = 5$ spins evolve under

$$H(u) = \sum_{i<j} J_{ij} X_i X_j + h \sum_i Z_i + h(1+u) \sum_i X_i, \tag{1}$$

with $h = 1/2$ and couplings J_{ij} drawn uniformly from $[-1, 1]$; the transverse field thus runs from h to $2h$. The environment adds a Lindblad dissipator [32, 33],

$$\dot\rho = -i[H(u), \rho] + \sum_k \Big(L_k \rho L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho\} \Big). \tag{2}$$

We compare the jump families of Fig. 1b: local loss $L_i = \sqrt{\gamma}\,\sigma_i^-$; one shared channel $L = \sqrt{\gamma}\sum_i c_i \sigma_i^-$ with equal phases $c_i = 1$ or mixed phases $c = (1, 1, 1, -1, -1)$; two shared channels on three and two spins; pairwise loss $\sigma_i^- \sigma_j^-$; local loss with exchange $\sigma_i^- \sigma_j^+$, which carry 40% and 60% of the coupling; local loss and gain with rates in the ratio $1 : 0.3$; and local dephasing σ_i^z. Any positive mixture in the exchange and gain/loss families leaves a single stationary state, as local loss does. Every family has the same total coupling $\operatorname{Tr} K = 80\gamma$, where $K = \sum_k L_k^\dagger L_k$ (Methods). This fixes the total amount of leaking exactly: averaged over all modes of the reservoir, the decay rate is $\operatorname{Tr} K/2^N$ for every family and every Hamiltonian. What differs is how the leaking is distributed.

The decisive difference is what the environment cannot reach. For local loss only the ground state is left alone. A shared channel annihilates a whole subspace of dark states, $\binom{N}{\lfloor N/2 \rfloor}$ of them on N spins. With five spins it has ten, and the dissipator leaves all 100 operators on them unchanged. Pairwise loss and the split channel leave six dark states, exchange and gain/loss a single stationary state, and dephasing leaves the 32 populations of the spin basis but erases every coherence (Fig. 1b).

A linear readout of the 45 single-spin and same-axis two-spin Pauli expectations reconstructs each past input. Its recall $C_\tau \in [0, 1]$ is the squared correlation between the estimate and the input τ steps back on held-out data, and the memory $\sum_\tau C_\tau$ sums it over delays up to 300. All expectation values are exact; there is no shot noise.

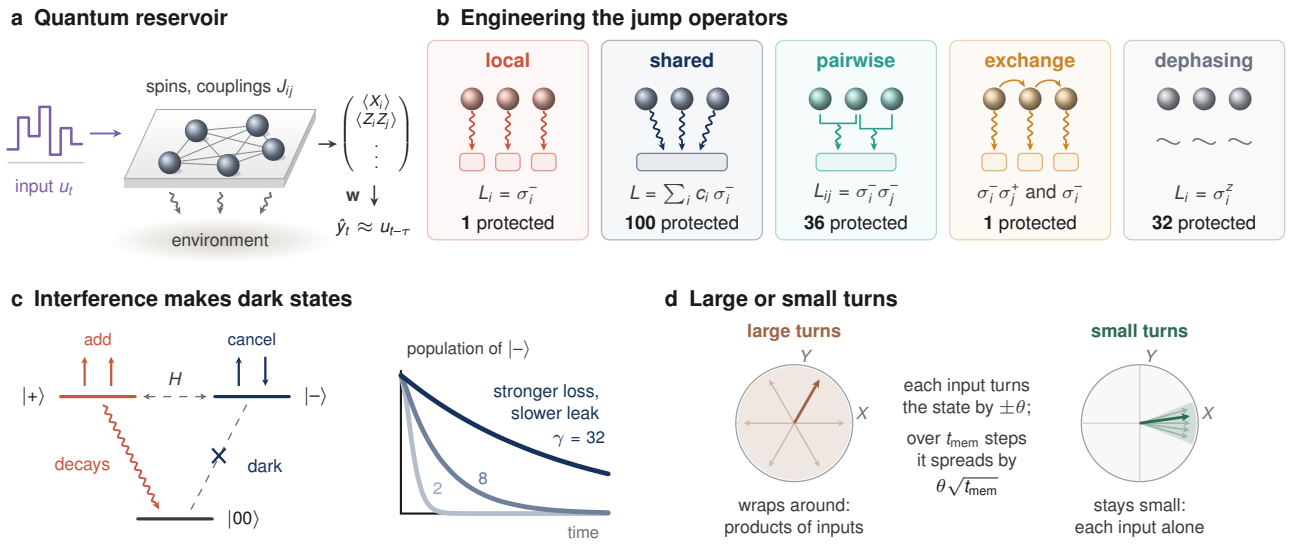


Figure 1: **Engineering the jump operators of a quantum reservoir.** **a** Inputs drive interacting spins coupled to an environment; a linear readout recalls past inputs. **b** Jump families at equal total coupling $\mathrm{Tr}\sum_k L_k^\dagger L_k$, with the number of operators each leaves untouched (protected) at five spins. **c** Two spins sharing one channel. The decay amplitudes of $|10\rangle$ and $|01\rangle$ add for $|+\rangle$, which decays, and cancel for $|-\rangle$, which is dark. The Hamiltonian H mixes the two; strong loss turns this into a slow leak, so the dark state survives longer the stronger the loss (inset: coupling g between $|+\rangle$ and $|-\rangle$, $\gamma/g = 2$, 8, 32). **d** Each input turns the stored state by $\pm\theta$, so over the memory time its direction spreads by about $\theta\sqrt{t_{\text{mem}}}$. Small turns keep the spread small and the readout sees each input on its own; large turns let it wrap around and the readout sees only products of inputs.

2.2 More loss, longer memory

Figure 2 shows the surprise. At weak dissipation all families behave alike, as the equal coupling suggests (Fig. 2a,b). As γ grows they split in opposite directions. With local loss, recall retreats to ever shorter delays, until at $\gamma = 64$ only the last input is left (Fig. 2a), and the memory falls steadily (Fig. 2b). With a shared channel the memory dips and then, beyond $\gamma \approx 1$, doubles; at $\gamma = 64$ the recall is still 0.25 fifty steps back and 0.07 a hundred steps back. Exchange and gain/loss follow local loss, and the split channel follows the shared one. Pairwise loss and mixed phases recover after a dip, but only to their weak-dissipation level, and at stronger loss their memory falls again; Sec. 2.5 explains why.

The time to forget makes the split quantitative (Fig. 2c). We compute it from the slow modes of the input-averaged step, the modes that the loss alone leaves untouched. Without protected operators it falls as $1/\gamma$. With dark states it grows with γ: in proportion to γ when the input cannot reach the dark states directly (one shared channel, two channels), where γ times the mean slow decay rate approaches 2.9 for one shared channel, on average over the reservoirs. When the input can reach them (mixed phases, pairwise loss), it scrambles the dark states itself, and the time to forget levels off, for pairwise loss only at larger γ (Methods). Stronger loss slows forgetting only when the jump operators leave dark states.

Dephasing shows what protection alone does not provide. Its 32 protected populations also forget ever more slowly, yet its memory is zero at every γ: dephasing and the unitary input both leave the maximally mixed state unchanged, and it is the unique stationary state for every input, so the reservoir relaxes to it whatever the input. Protection is useful only for what the input can write into it. With exact features, recall counts what is linearly recoverable however small the signal; we return to the size of the signal in Sec. 2.6.

2.3 Dark states that cannot leak

Why does more loss help a shared channel? A strong channel acts like constant checking of whether an excitation has left, and such checking holds the state where the answer is no [34, 28] (Fig. 1c). At strong loss the reservoir follows, in a first approximation, its Hamiltonian projected onto the operators that the dissipator leaves alone, which turns the state but cannot forget. Forget-

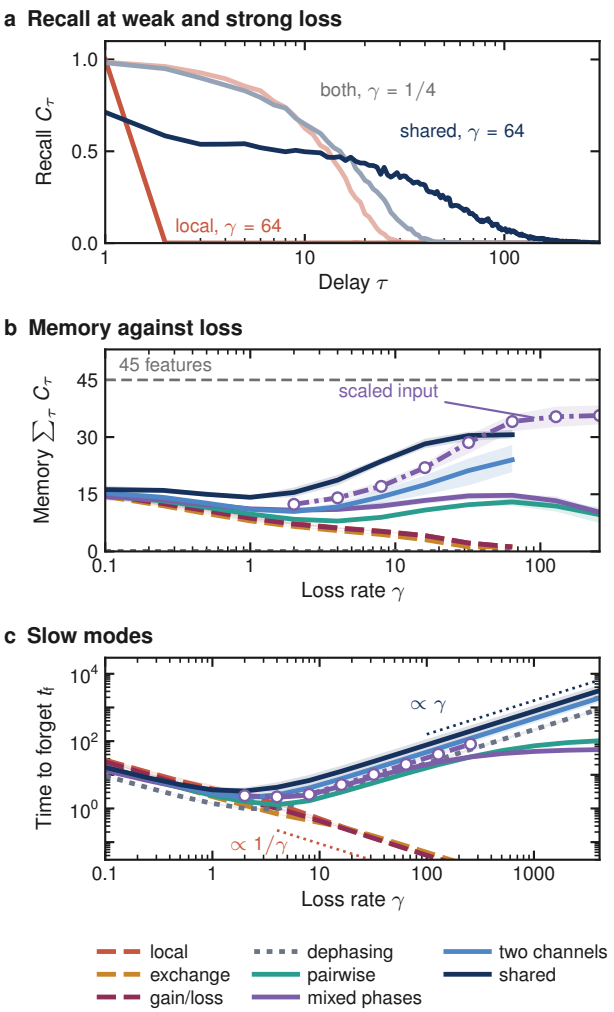


Figure 2: **Same total leaking, opposite outcomes.** Five spins at equal total coupling; mean and 16th–84th percentiles over eight reservoirs. **a** Recall of the input τ steps back at weak ($\gamma = 1/4$, light) and strong ($\gamma = 64$, dark) loss. **b** Memory at a fixed input amplitude; one and two shared channels stop at $\gamma = 64$, where their recall outgrows the 300 delays summed (Methods). Dash-dotted with circles: mixed phases with the input amplitude scaled as $\min(1, \gamma^{-1/2})$; dashed: the limit set by the 45 features. **c** Time to forget, from the slow modes of the input-averaged step. Dephasing protects populations into which the input never writes, so its memory in **b** is zero.

ting enters only in the next correction, when the Hamiltonian briefly lifts a dark state into a bright one that decays [29, 35]:

$$\mathcal{G}(u) = \mathcal{P}\mathcal{H}(u)\mathcal{P} - \frac{1}{\gamma}\,\mathcal{P}\mathcal{H}(u)\mathcal{S}\mathcal{H}(u)\mathcal{P}, \qquad (3)$$

with $\mathcal{H} = -i[H, \cdot]$, $\mathcal{P}$ the projector onto the protected operators and $\mathcal{S}$ the inverse of the unit-rate dissipator on the rest (Methods). This is a rotation plus a leak of size $1/\gamma$: an orthogonal map at the edge of stability, the classic recipe for long memory in a network [36]. Equation (3) holds for any number of spins, so at strong loss the leak falls as $1/\gamma$ at every size; the size sets only its coefficient, which we compute for three to six spins (Methods), and how strong the loss must be for this regime to begin.

Three tests confirm the picture; Fig. 3a counts the protected operators of every family. At strong loss, 99 eigenvalues of the averaged step approach the unit circle, one for each operator on the dark subspace apart from the stationary state; with local loss they stay near zero (Fig. 3b). The ten-state model of Eq. (3), with nothing fitted, reproduces the recall of the full reservoir from $\gamma \approx 4$ on, within a few percent (Fig. 3c). Erasing the dark coordinates after every step destroys all memory beyond a few steps, while erasing a rotated copy of the same dimension leaves most of it (Fig. 3d). The dark states hold the memory.

2.4 A long memory blends its inputs

Lasting longer is not the same as staying readable. Every input nudges the stored state a little, and the nudge depends on the input (Fig. 1d). Over its memory time the stored direction takes a random walk of steps $\pm\theta$ and spreads by about $\theta\sqrt{t_{\mathrm{mem}}}$. While the spread is small, the measured projection is linear in each input. Once it reaches about a radian, the projection wraps around: the record of an old input survives only multiplied by the inputs that followed it. A linear readout of a few observables recovers single inputs; it cannot turn such products back into them. We call the number of distinct inputs that a stored function involves its order, and we resolve the memory by order [37, 38] (Methods).

The simplest protected reservoir is reset to a fixed state with a small probability p in each step and is otherwise turned by an input-dependent unitary. The variance of a stored signal then decays over $t_{\mathrm{mem}} = 1/(2p)$ steps. Write the step in traceless coordinates as an orthogonal map $O(u)$ and call $s = 1 - \|\mathbb{E}O\|_F^2/n$ the mixing fraction of one input. It measures how hard one input nudges the stored state: the mean squared fraction of a randomly oriented state that one input moves away from the average turn. For small turns it grows as the square of the input amplitude.

Theorem 1 (Mixing law)**.** *Let a reservoir be*

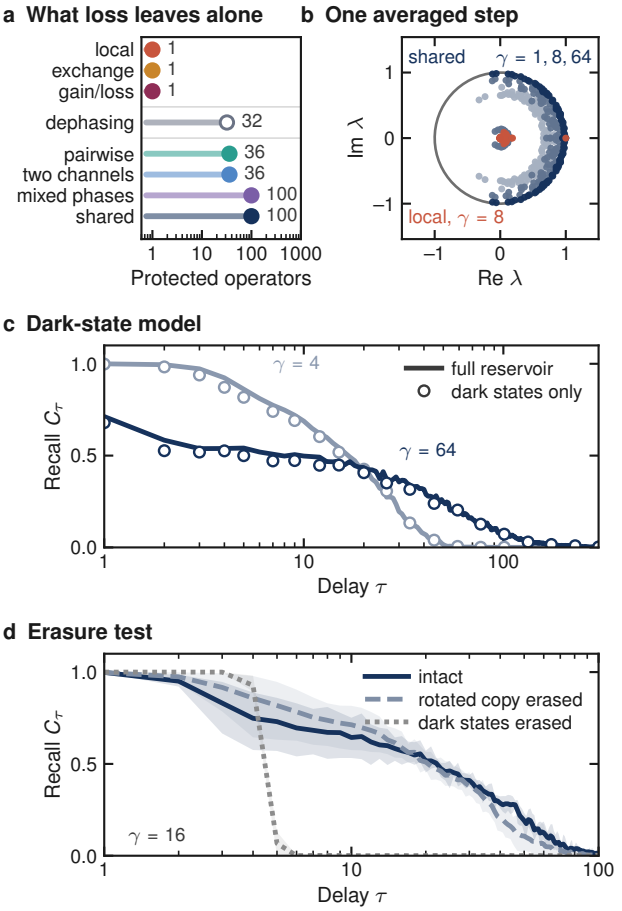


Figure 3: **Dark states hold the memory.** **a** Number of operators that each unit-rate dissipator leaves unchanged, at five spins. Rules separate the families without a dark state, dephasing, which protects populations only (open circle), and the families with a dark subspace. **b** Eigenvalues of the input-averaged step of one reservoir: one shared channel at $\gamma = 1$, 8 and 64 (light to dark) and local loss at $\gamma = 8$. **c** Recall of the full reservoir (lines) and of the dark-state model of Eq. (3) (circles), with nothing fitted; mean of eight reservoirs. **d** Recall at $\gamma = 16$ when the dark coordinates, or a copy of the dark subspace rotated by random single-spin unitaries, are erased after every step; mean and 16th–84th percentiles over eight reservoirs.

reset with probability p per step and otherwise turned by input-dependent rotations that mix the stored state and write a signal (conditions in Appendix A). Draw the order D_p of a stored function with probability equal to its share of the memory. As $p \to 0$, pD_p becomes exponentially distributed, with mean

$$\mathbb{E}D_p = s\, t_{\mathrm{mem}} + O(1).$$

In words, a memory blends on average $s\, t_{\mathrm{mem}}$ inputs: how hard one input nudges it times how long it lasts. Halving the reset probability doubles the typical order, and any fixed set of orders carries a vanishing share of the memory although its total stays constant (Appendix A). In a protected reservoir the leakage plays the role of p, so at a fixed input amplitude a longer memory blends more and more inputs.

Blending is not only a cost. Products of inputs are the nonlinear memory that tasks such as parity need [37, 39], and the law says how many inputs a reservoir combines. In the first version of this study the equal-phase shared channel scored below local loss on parity and closed-loop forecasting [31] (Appendix D). This is consistent with its input having no matrix element between its dark states (Sec. 2.5): at $\gamma = 1$ its readout recovers half as much capacity of polynomial degree three and four as that of local loss, while with mixed phases, where the input reaches the dark states directly, it recovers as much as with local loss (Methods).

2.5 Write more gently

The law also says how to keep a long memory readable: write more gently as it grows. A single qubit that relaxes towards $+X$ at rate ϵ, is turned about Z by $\pm\theta$ per input and is read along Y shows the whole trade-off in closed form (Fig. 4a, Appendix B). Everything depends on θ^2/ϵ, the squared turn per memory time. For large values the inputs blend, and they also scramble the polarization that each new input tilts, so the stored signal shrinks. For small values each input writes too little. The stored signal is largest at $\theta^2/\epsilon = 2/\sqrt{3}$. Both results follow one rule: keep the mixing fraction times the memory time of order one. With $s \propto \theta^2$ and $t_{\mathrm{mem}} \propto 1/\epsilon$ this means $\theta^2 \propto \epsilon$: the input amplitude must fall as one over the square root of the memory time, so a memory four times longer needs inputs half as strong, with a prefactor set by the reservoir. We call this gentle writing.

The input must also reach the dark states directly, and the jump operators decide whether it does. The dark states of a shared channel $L \propto \sum_i c_i \sigma_i^-$ are annihilated by $\sum_i c_i \sigma_i^-$. An input field $\sum_i c_i X_i$ along the same phase vector is the sum of this operator and its adjoint, so it has no matrix element between dark states. It writes only through brief excursions into bright states, and its effect falls as $1/\gamma$, while an input pattern across the phase vector reaches the dark

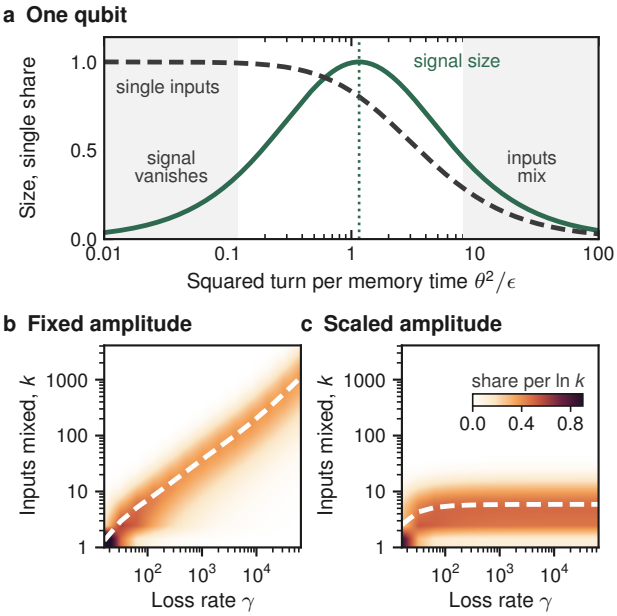


Figure 4: **Writing more gently keeps a long memory readable. a** One qubit relaxing at rate ϵ and turned by $\pm\theta$ per input, in the limit $\epsilon \to 0$ at fixed θ^2/ϵ: size of the stored signal (normalized to its maximum at $\theta^2/\epsilon = 2/\sqrt{3}$, dotted) and share of single past inputs (dashed). **b,c** Three-spin reservoir with one shared channel of mixed phases (Hamiltonian H3): share of the memory of X_1 carried by products of k inputs, per unit $\ln k$, for a fixed input amplitude (b) and for an amplitude scaled as $8/\sqrt{\gamma}$ (c); dashed: mean order; one colour scale for both.

states directly (Methods). This explains the families of Fig. 2. The uniform input of Eq. (1) lies along the phases of the equal-phase channel and of each split channel, so it reaches the dark states only through bright states: their memory grows with the loss, but the signal behind it shrinks. For mixed phases and for pairwise loss the same input reaches the dark states directly, and at a fixed amplitude their records blend with later inputs, so the memory of single inputs stays near its weak-dissipation level and falls at the strongest loss. Gentle writing removes this limit. With the input amplitude scaled as $\gamma^{-1/2}$, the memory of the mixed-phase channel grows with the loss like that of the shared channel, its time to forget grows in proportion to γ again, and at strong loss the signal in each feature keeps a constant size (dash-dotted in Fig. 2b,c; Methods).

A three-spin reservoir with one shared channel of mixed phases shows both regimes, and there we prove them (Methods). With a fixed input amplitude the order of its memory grows in proportion to γ and single inputs lose their share (Fig. 4b), while the signal in a measured spin shrinks as $1/\gamma$. With the amplitude scaled as $8/\sqrt{\gamma}$, which exceeds the fixed amplitude below $\gamma = 64$ and falls below it above, the order stays small (Fig. 4c), single inputs keep a finite share, the signal stays finite, and the range of delays that holds 90% of the single-input recall grows in proportion to γ. Computer-assisted bounds prove both limits for four Hamiltonians and for neighbourhoods of their couplings (Appendix C).

Coherence between dark states carries this memory. Coherence is known to shape the performance of spin reservoirs [40]; here we isolate its role. We return the spins to the dark subspace after every input and compare two arms that differ only in a dephasing of the dark energy levels. The populations follow the same slow dynamics in both arms, but without coherence the stored signal collapses as $1/\gamma^2$ (Methods). A new input first turns a population into a coherence, and erasing the coherences deletes each record as soon as it is written (Appendix C).

2.6 Longer but thinner

Protection with gentle writing makes the memory longer, not larger. The memory cannot exceed the number of features, 45 here (Fig. 2b, dashed), and one observable holds at most one unit. As the time to forget keeps growing, each delay is therefore recalled more faintly. With five spins and mixed phases, the gently written memory levels off near 36, below the ceiling of 45, while the recall of the last input falls from 0.95 at $\gamma = 64$ to 0.57 at $\gamma = 256$. With three spins the total recall of single inputs stays fixed while its range grows as γ, so each delay is recalled as $1/\gamma$.

The signal in each feature is small. Exact expectation values stand for averages over many repeated runs. Three measurement settings, with all spins read along X, Y or Z, give all 45 features, and a feature whose signal has variance v needs roughly $1/v$ shots per setting and step. With gentle writing v stays constant as the loss grows, near 5×10^{-6} for an average feature of the five-spin reservoir, sixteen times less than at a fixed amplitude; this asks for about 2×10^5 shots per setting and step; with the uniform input of a shared channel it shrinks, so the shots needed grow too (Methods). In the first version of this study, finite sampling hid the advantage of the shared channel below about 2×10^5 preparations

per step (Appendix D).

Ordinary noise caps the gain. Local loss at rate γ_{loc} limits the memory time to about $1/(b\gamma_{\mathrm{loc}}\Delta t)$ steps, with $b \approx 1.41$, so for these five-spin reservoirs a memory of T steps needs collective loss at least about $4T^2$ times stronger than local loss. Local dephasing acts the same way, with $b_\phi \approx 4.07$ and a ratio of about $12T^2$ (Methods). In a waveguide, a dark state of four transmons has been shown to live more than a hundred times longer than a single qubit decaying into the guide [41]; read as $\gamma/\gamma_{\mathrm{loc}}$ in this bound, a ratio of 100 allows a memory of about five steps, and every factor of four doubles it.

3 Discussion

Engineering the jump operators turns dissipation from a clock into a design layer. At equal total coupling, what the environment cannot reach decides whether stronger dissipation speeds up forgetting or slows it down, and what the input can reach decides whether the slower forgetting becomes memory. Local loss, exchange and gain/loss leave a single stationary state, so more loss means faster forgetting. Shared and pairwise loss leave dark subspaces, so more loss means slower forgetting as long as the input does not scramble them itself. Dephasing protects populations that the input never reaches.

Three design rules follow for the reservoirs we studied: choose jump operators with a dark subspace, so that the loss rate sets the memory time; let the input reach this subspace directly, which the relative phases of a shared channel and of the input decide; and write more gently as the memory grows, keeping the mixing fraction of one input times the memory time of order one. The last two rules form one trade-off: direct access gives a large signal, and gentle writing keeps it from blending.

The scope is clear. The mixing law assumes rotation-like inputs, slow resetting and a single way in which stored signals spread; with two independent sectors one observable can show two order scales (Appendix A). The spin results concern reservoirs of five and three spins with specified Hamiltonians, and memory means what a linear readout of exact expectation values recovers; Sec. 2.6 gives the cost of reading it from finite measurements and the limit set by ordinary noise. A generic common bath need not realize the engineered channels; shared resonators, waveguides and structured couplers show how they can be built [42, 43, 44, 45]. How protection combines with non-Markovian memory [13] is an open question.

How a quantum system leaks decides not only how fast it forgets, but what it keeps and in what form. The environment stops being only the place where information is lost and becomes a tool for deciding what is kept.

Acknowledgments

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Data availability

The code, the data behind every figure and the interval certificates of the three-spin limits are available at `https://github.com/eybmits/stronger-dissipation-longer-memory`; the certificates are also attached to this PDF.

Author contributions

M.B. contributed to conception, methodology, software, analysis, visualization and writing. I.F. and J.W. contributed to methodology, interpretation and editing. C.L.-P. and J.S. contributed to supervision and conceptual development. OpenAI Codex assisted with drafting, literature checks, derivations, code, numerical experiments, figures and verification software; it also wrote the independent finite-rate implementation of Appendix C as a separate program that shares no routine with the certificate programs. All AI-assisted material was independently reviewed and verified by the authors, who take full responsibility for the scientific content and final manuscript.

4 Methods

4.1 Five-spin simulations

Each reservoir draws its couplings and its input stream from one seed, and every family uses the same eight seeds. Equation (2) is integrated in the

1024-dimensional basis of Pauli operators. The input is constant within a step, and the step map $E(u) = e^{\Delta t \mathcal{L}(u)}$ is evaluated by Chebyshev interpolation in u at ten nodes. After a washout of $\max(800, 60\gamma)$ steps we record the 45 features for 2000 training, 400 validation and 1600 test inputs. The readout is a least-squares fit with an intercept on the training and validation inputs, and C_τ is the squared correlation of its estimate with $u_{t-\tau}$ on the test inputs. All features are exact expectation values. Runs at $\gamma = 128$ and 256 use the same seeds. There the recall of one and two shared channels at a fixed amplitude reaches past the 300 delays we sum over (the time to forget of the shared channel is about 200 steps at $\gamma = 256$), so Fig. 2b shows them up to $\gamma = 64$. The recall of mixed phases and pairwise loss ends well within the window, and their memory falls to about 10 at $\gamma = 256$. Capacities by polynomial degree follow Ref. [37]: products of Legendre polynomials in the past inputs $2u - 1$, up to total degree four within finite delay windows, fitted by the same features and kept above a shuffled-input noise floor. At $\gamma = 1$ degrees three and four carry 8.2 units with one shared channel, 15.9 with mixed phases and 16.4 with local loss.

For the scaled input we replace $h(1 + u)$ in Eq. (1) by $h[\frac{3}{2} + \eta(u - \frac{1}{2})]$ with $\eta = \min(1, \gamma^{-1/2})$, which keeps the mean field and scales the input amplitude. With mixed phases the variance of a feature over the test inputs, averaged over the 45 features, then stays near 5×10^{-6} from $\gamma = 16$ to 256, and from $\gamma = 32$ on the largest is about 2×10^{-5}. Meanwhile the recall of the last input falls from 0.95 at $\gamma = 64$ to 0.57 at $\gamma = 256$ and spreads to longer delays; beyond delay 300 it would add about 1% of the memory on average. With $\eta = \min(1, 2\gamma^{-1/2})$ the features vary 3.5 times more, and the memory is smaller, 29 instead of 34 at $\gamma = 64$, consistent with stronger mixing. At a fixed amplitude the mean feature variance of the equal-phase shared channel falls from 2×10^{-4} at $\gamma = 1$ to 4×10^{-7} at $\gamma = 64$ and 3×10^{-8} at $\gamma = 256$, while that of mixed phases stays near 8×10^{-5} from $\gamma = 16$ on.

4.2 Jump families and equal coupling

Table 1 lists the jump operators. Their weights give every family the same total coupling $\operatorname{Tr} K = \sum_k \operatorname{Tr} L_k^\dagger L_k = N 2^{N-1} \gamma = 80\gamma$; the mixtures for exchange and gain/loss follow the first version of this study. For a traceless jump the superoperator trace of the dissipator is $\operatorname{Tr} \mathcal{D}[L] = |\operatorname{Tr} L|^2 - 2^N \operatorname{Tr} L^\dagger L = -2^N \operatorname{Tr} L^\dagger L$, while the commutator with any Hamiltonian is traceless. The 4^N eigenvalues λ of the generator therefore obey

$$\frac{1}{4^N} \sum \operatorname{Re} \lambda = -\frac{\operatorname{Tr} K}{2^N} \tag{4}$$

for every family and every Hamiltonian. The protected operators form the kernel of the unit-rate dissipator, which we compute numerically (Fig. 3a).

4.3 Slow modes

The input-averaged step $\bar{E} = \mathbb{E}_u E(u)$ is computed by six-point Gauss–Legendre quadrature. Its eigenvalues λ_j give decay rates $\kappa_j = -\ln|\lambda_j|/\Delta t$. If the dissipator protects n operators, the slow modes are the $n - 1$ slowest decaying ones besides the stationary state, and the time to forget is $t_\mathrm{f} = 1/(\Delta t\, \bar{\kappa})$ with $\bar{\kappa}$ their mean rate. Without protected operators we use the slowest mode. For one shared channel $\gamma\bar{\kappa} = a = 2.945$ to four digits from $\gamma = 512$ to 16384, so t_f grows fourfold from $\gamma = 4096$ to 16384, while it changes from 56 to 58 steps with mixed phases and from 103 to 114 with pairwise loss. The same second-order calculation as in Eq. (3), which gives $a = 2.945$ at five spins, gives $a = 1.4$, 2.1 and 3.4 at three, four and six spins (mean over the same eight coupling seeds), so the leak coefficient grows only slowly with the size. Adding local loss $\sqrt{\gamma_\mathrm{loc}}\, \sigma_i^-$ gives $\bar{\kappa} = a/\gamma + b\gamma_\mathrm{loc}$. To first order in γ_loc, b is the trace of the added dissipator on the 99 slow dark modes divided by 99, $b = 140/99 \approx 1.414$. Local dephasing $\sqrt{\gamma_\phi}\, \sigma_i^z$ (the dephasing family of Table 1

Table 1: Jump families at $N = 5$, each with $\operatorname{Tr} K = 80\gamma$, and the number of operators that the dissipator leaves unchanged; $i < j$ for pairwise loss, $i \neq j$ for exchange and $c = (1, 1, 1, -1, -1)$.

family	jump operators	protected
local	$\sqrt{\gamma}\, \sigma_i^-$	1
shared	$\sqrt{\gamma} \sum_i \sigma_i^-$	100
mixed phases	$\sqrt{\gamma} \sum_i c_i \sigma_i^-$	100
two channels	$\sqrt{\gamma} \sum_{i \le 3} \sigma_i^-$, $\sqrt{\gamma} \sum_{i > 3} \sigma_i^-$	36
pairwise	$\sqrt{\gamma}\, \sigma_i^- \sigma_j^-$	36
exchange	$\sqrt{0.4\gamma}\, \sigma_i^-$, $\sqrt{0.3\gamma}\, \sigma_i^- \sigma_j^+$	1
gain/loss	$\sqrt{\gamma/1.3}\, \sigma_i^-$, $\sqrt{0.3\gamma/1.3}\, \sigma_i^+$	1
dephasing	$\sqrt{\gamma/2}\, \sigma_i^z$	32

has $\gamma_\phi = \gamma/2$) adds $b_\phi\gamma_\phi$ in the same way, with $b_\phi = 403.2/99 \approx 4.073$ to first order. For γ from 32 to 256 and noise rates from 0.001 to 0.1 the full spectra reproduce both coefficients to within 0.2%. A time to forget of at least T steps needs $a/\gamma + b\gamma_{\mathrm{loc}} \le 1/(\Delta t\, T)$, which is possible only if $\gamma/\gamma_{\mathrm{loc}} \ge 4ab\,\Delta t^2 T^2 \approx 4.2\,T^2$; for dephasing the bound is $4ab_\phi\Delta t^2T^2 \approx 12\,T^2$.

4.4 Dark-state model and erasure

Let $\mathcal{D}$ be the unit-rate dissipator, $\mathcal{P} = \lim_{t\to\infty} e^{t\mathcal{D}}$ the projector onto its kernel and $\mathcal{S} = (\mathcal{D}+\mathcal{P})^{-1} - \mathcal{P}$ the inverse on the decaying operators. Write $\mathcal{P} = RR^\flat$ with $R^\flat R = 1$, where R embeds the 100 dark coordinates α. Eliminating the decaying operators to second order gives Eq. (3) for α, and the full state is $(R - \mathcal{S}\mathcal{H}(u)R/\gamma)\,\alpha$, which we feed to the same readout. Nothing is fitted. For the erasure test we replace the dark coordinates after every step by their time average, $r \mapsto r - RR^\flat(r - \bar r)$; the control erases the dark subspace rotated by a random product of single-spin unitaries. Recall is then measured at delays 1 to 100 with 800 training, 200 validation and 800 test inputs. The write per input is $2\|R^\flat[E(\frac{3}{4}) - E(\frac{1}{4})]r_*\|$ at the stationary state r_* of the equal-phase channel, for the input patterns $(1,\dots,1)$ and $\sqrt{2}\cos(2\pi i/5)$ multiplying X_i. Along the phases it falls as $1/\gamma$, to 0.003 at $\gamma = 128$; across them it stays of order one, 0.18 at $\gamma = 128$.

4.5 Order of a memory

For independent signs $u_t = \pm 1$ the products $\prod_{j\in J} u_{t-j}$ over finite sets J of delays are orthonormal and complete; their order is $|J|$. For continuous inputs one uses orthogonal polynomials in each input, and any centred function of one input has order one. Quantum channels act affinely on traceless coordinates [46, 47], $r_t = A(u_t)r_{t-1} + b(u_t)$. Let $\bar A = \mathbb{E}A$, m the stationary mean and $\xi = A(u)m + b(u) - m$ the innovation, with covariance V. With $\mathcal{B}X = \bar A X \bar A^{\mathsf T}$, $\mathcal{T}X = \mathbb{E}[AXA^{\mathsf T}]$ and $\mathcal{K} = \mathcal{T} - \mathcal{B}$, the covariance G_k carried by products of order k obeys

$$G_1 = (1-\mathcal{B})^{-1}V, \qquad G_{k+1} = (1-\mathcal{B})^{-1}\mathcal{K}G_k, \tag{5}$$

because the mean step moves a stored product without adding a time, the fluctuating step multiplies it by the current input, and the innovation adds the current input alone. The G_k add up to the stationary covariance Σ. A readout of features $y = Mx$ explains the share $C_k = \mathrm{Tr}[(M\Sigma M^{\mathsf T})^{+} M G_k M^{\mathsf T}]$ of order k, and these shares add up to the rank of the readout [37]. For one observable a, $C_k = a^{\mathsf T}G_k a / a^{\mathsf T}\Sigma a$, and the share of single past inputs is $a^{\mathsf T}\bar A G_1 \bar A^{\mathsf T} a / a^{\mathsf T}\Sigma a$.

4.6 Three-spin computations

The three spins evolve under

$$H_\eta(u) = \sum_{i<j} J_{ij}X_iX_j + \tfrac{1}{2}\sum_i Z_i + \tfrac{3+\eta u}{4}\sum_i X_i \tag{6}$$

with one jump $\sqrt{\gamma}\,(\sigma_1^- - \sigma_2^- - \sigma_3^-)$, steps of $\Delta t = 1/2$ and signs $u = \pm 1$, at $\eta = 1$ (fixed amplitude) or $\eta = 8/\sqrt{\gamma}$ (scaled amplitude). The dark subspace has three states. The couplings of the four Hamiltonians H1–H4 are drawn from fixed seeds. For each loss rate we exponentiate the full generator for both signs, solve the stationary moment equations and evaluate Eq. (5) for X_1 until 99.99% of its memory is accounted for; there is no sampling. The range τ_{90} is the smallest delay beyond which less than a tenth of the recall remains. For the coherence test we apply the loss-only projector $\mathcal{P}$ after every step; one arm keeps the recovered state, the other also dephases it in the energy basis of the projected Hamiltonian. From $\gamma = 2^8$ to 2^{18} the variance of X_1 stays between 10^{-4} and 2×10^{-3} when the coherences are kept and falls as γ^{-2}, by six orders of magnitude, when they are erased; at a fixed input amplitude it falls as $1/\gamma$.

A The mixing law

A.1 Precise statement

On n traceless coordinates let the step be $r_t = (1-p)O(u_t)r_{t-1} + pb$, where independent, identically distributed inputs select orthogonal maps $O(u)$ and b represents the reset state. Assume that the mean map $A = \mathbb{E}O$ has spectral radius below one, that the only symmetric matrices fixed by $X \mapsto \mathbb{E}[OXO^{\mathsf T}]$ are multiples of the identity, and that the input writes a signal, $\mathbb{E}\|(O-A)(1-A)^{-1}b\|^2 > 0$. For a fixed linear readout of positive rank, draw the order D_p with probability equal to its share of the ideal memory. Then $pD_p \to \mathrm{Exp}(2/s)$ in distribution with $s = 1 - \|A\|_F^2/n$, and $\mathbb{E}D_p = s/(2p) + O(1)$.

Idea. One input writes a variance of order p^2. A covariance shape that survives for $1/(2p)$ steps collects it to order p. We identify that shape and then count, with the recursion of Eq. (5), how many inputs it contains.

A.2 Generating function

The order transfer $\mathcal{F} = (1-\mathcal{B})^{-1}\mathcal{K}$ is a positive map on covariance matrices, and $(1-\mathcal{F})^{-1} = (1-\mathcal{T})^{-1}(1-\mathcal{B}) = 1 + (1-\mathcal{T})^{-1}\mathcal{K}$ is positive, so its spectral radius is below one. Summing Eq. (5),

$$\begin{aligned} \sum_{k\geq 1} z^k G_k &= z(1-z\mathcal{F})^{-1} G_1, \\ \sum_{k\geq 1} k G_k &= (1-\mathcal{F})^{-1}\Sigma, \end{aligned} \tag{7}$$

and $\mathcal{F}^K\Sigma$ is the covariance above order K.

A.3 The surviving covariance

Let $r = 1-p$, $A = \mathbb{E}O$, $\mathcal{B}_0 X = AXA^\mathsf{T}$ and $\mathcal{T}_0 X = \mathbb{E}[OXO^\mathsf{T}]$. Every step contracts differences by r, so there is a unique stationary state. Its mean is $m_p = p(1-rA)^{-1}b = p\mu + O(p^2)$ with $\mu = (1-A)^{-1}b$, and the innovation $\xi_p = r(O-A)m_p$ has covariance $V_p = p^2 V_0 + O(p^3)$ with $v = \operatorname{Tr} V_0 = \|\mu\|^2 - \|A\mu\|^2 > 0$ by the writing assumption. On symmetric matrices $\mathcal{T}_0$ preserves the trace and the identity and contracts the Frobenius norm; by assumption the identity is its only fixed point, and power boundedness excludes a Jordan block. Its spectral projector is $X \mapsto 1 \operatorname{Tr} X/n$, and all other directions have a bounded resolvent near one. Hence

$$\Sigma_p = (1-r^2\mathcal{T}_0)^{-1} V_p = \frac{pv}{2n}\, 1 + O(p^2). \tag{8}$$

The mean map is mixing, so $(1-r^2\mathcal{B}_0)^{-1}$ stays bounded and every fixed order has $G_k = O(p^2)$. Since $\Sigma_y^+ = O(p^{-1})$ on its support, any fixed set of orders carries a share $O(p)$, while the shares add up to the rank q of the readout.

A.4 The number of inputs

At $p = 0$ the transfer $\mathcal{F}_0 = (1-\mathcal{B}_0)^{-1}(\mathcal{T}_0 - \mathcal{B}_0)$ fixes the identity; it is positive and power bounded, and its unit eigenvalue is simple. Its left eigenfunctional is

$$\ell(X) = \operatorname{Tr}[(1-A^\mathsf{T}A)X], \quad \ell(1) = ns > 0, \tag{9}$$

because $\ell = \operatorname{Tr}\circ(1-\mathcal{B}_0)$ and $\operatorname{Tr}\circ\mathcal{T}_0 = \operatorname{Tr}$. Differentiating $(1-r^2\mathcal{B}_0)\mathcal{F}_p = r^2(\mathcal{T}_0 - \mathcal{B}_0)$ at $p = 0$ gives $(1-\mathcal{B}_0)\mathcal{F}_0' 1 = -2\cdot 1$, so the leading eigenvalue is $1 - \beta p + O(p^2)$ with $\beta = 2n/\ell(1) = 2/s$. Insert $z = e^{-\lambda p}$ into Eq. (7): for fixed $\lambda \geq 0$ only this eigenvalue gives a divergent resolvent term, and with $\ell(G_1) = p^2 v + O(p^3)$,

$$\sum_{k\geq 1} e^{-\lambda p k} G_k = \frac{pv}{\ell(1)(\lambda+\beta)}\, 1 + O(p^2). \tag{10}$$

Divided by Eq. (8) this is the Laplace transform $\beta/(\lambda+\beta)$ of the exponential law. The second identity in Eq. (7) gives the order-weighted covariance $\Sigma_p/(\beta p) + O(p)$, hence $\mathbb{E}D_p = 1/(\beta p) + O(1) = s\, t_{\mathrm{mem}} + O(1)$.

A.5 The mixing fraction

For a unit vector x drawn uniformly, $\mathbb{E}\|(O-A)x\|^2 = \operatorname{Tr}\mathcal{K}_0(1)/n = s$: one input moves a fraction s of a stored signal away from the average turn, and this fraction is passed to the next order. For $O = e^{u\theta G}$ with $u = \pm 1$ and skew-symmetric G with eigenvalues $i\lambda_j$, the mean map is $\cosh\theta G$ and $s = n^{-1}\sum_j \sin^2\theta\lambda_j$, which grows as θ^2.

A.6 Two sectors, two laws

The assumption of a single fixed covariance matters. Let a classical flag select one of two qubit rotations about Z, by $\pm\pi/6$ or $\pm\pi/3$, and let the reset state be block diagonal. Each sector obeys the law separately, and a single Y readout sees the mixture $\Pr(pD_p > x) \to \frac{1}{2}e^{-8x} + \frac{1}{2}e^{-8x/3}$.

A.7 Physical channels

A physical channel does not reset to a prescribed state, but the argument above only needs a single slow covariance direction. We state the general version that Appendix C uses and prove it along the same four steps.

Setting. Work on invariant reachable coordinates and translate a common limiting fixed point to zero; for the spin reservoir of Appendix C these are the dark coordinates, which the bright part follows at order ϵ [Eq. (18)]. The step is $r_t = A_{u_t}(\epsilon) r_{t-1} + b_{u_t}(\epsilon)$ with a small leakage ϵ, and we write $\bar{A}_\epsilon = \mathbb{E}A_u(\epsilon)$, $\mathcal{B}_\epsilon X = \bar{A}_\epsilon X \bar{A}_\epsilon^\mathsf{T}$, $\mathcal{T}_\epsilon X = \mathbb{E}[A_u X A_u^\mathsf{T}]$ and $\mathcal{K}_\epsilon = \mathcal{T}_\epsilon - \mathcal{B}_\epsilon$. Assume: (i) $A_u(\epsilon) = A_{u,0} + \epsilon A_{u,1} + O(\epsilon^2)$ and $b_u(\epsilon) = \epsilon b_{u,1} + O(\epsilon^2)$ in mean square; (ii) the

mean step is mixing at $\epsilon = 0$, that is, $\bar{A}_0$ has spectral radius below one; (iii) the channel is mean-square stable for small $\epsilon > 0$; (iv) $\mathcal{T}_\epsilon$ has a simple eigenvalue $1 - \kappa\epsilon + O(\epsilon^2)$ with $\kappa > 0$, all its other eigenvalues stay a fixed distance away from one, and at $\epsilon = 0$ the eigenvalue has right eigenmatrix $R \succeq 0$ and a left eigenfunctional ℓ of $\mathcal{T}_0$ that is nonnegative on positive matrices, with $\ell(R) = 1$; (v) the input writes along it, $\ell(V_*) > 0$, where V_* is defined in step 1.

Step 1: the innovation. The stationary mean solves $m = \bar{A}_\epsilon m + \mathbb{E}b_u(\epsilon)$. By (ii) it is $m_\epsilon = \epsilon m_1 + O(\epsilon^2)$ with $m_1 = (1 - \bar{A}_0)^{-1}\mathbb{E}b_{u,1}$. The innovation is therefore $\xi_u = \epsilon z_u + O(\epsilon^2)$ with $z_u = A_{u,0}m_1 + b_{u,1} - m_1$, and its covariance is $V_\epsilon = \epsilon^2 V_* + O(\epsilon^3)$ with $V_* = \mathbb{E}[z_u z_u^\mathsf{T}]$; this is the order-$p^2$ writing of Appendix A.3.

Step 2: the surviving covariance. Let $\mathcal{P}_\epsilon$ be the spectral projector of the simple eigenvalue in (iv); it depends smoothly on ϵ and $\mathcal{P}_0 X = R\,\ell(X)$. On the complement $(1 - \mathcal{T}_\epsilon)^{-1}$ stays bounded, while on the slow direction it divides by $\kappa\epsilon + O(\epsilon^2)$. Since $\mathcal{P}_\epsilon V_\epsilon = \epsilon^2 \ell(V_*)R + O(\epsilon^3)$,

$$\begin{aligned} \Sigma_\epsilon &= (1 - \mathcal{T}_\epsilon)^{-1} V_\epsilon = \epsilon\, C_* + O(\epsilon^2), \\ C_* &= \frac{\ell(V_*)}{\kappa}\, R. \end{aligned} \tag{11}$$

Step 3: every fixed order is small. By (ii) the spectral radius of $\mathcal{B}_\epsilon$ stays below one, so $(1 - \mathcal{B}_\epsilon)^{-1}$ is bounded. Equation (5) then gives $G_1 = O(\epsilon^2)$ and, since $\mathcal{K}_\epsilon$ is bounded, $G_k = O(\epsilon^2)$ for every fixed k. Its share of the observed variance of a readout M tends to zero whenever $\mathrm{Tr}(MC_*M^\mathsf{T}) > 0$. For the ideal memory of a readout with several features we also need MC_*M^T to have full rank on the measured support, so that $(M\Sigma_\epsilon M^\mathsf{T})^+ = O(\epsilon^{-1})$; for a single observable this holds trivially.

Step 4: the number of inputs. At $\epsilon = 0$ the order transfer $\mathcal{F}_0 = (1 - \mathcal{B}_0)^{-1}\mathcal{K}_0$ has the same fixed points as $\mathcal{T}_0$, because $\mathcal{F}_0 X = X$ is equivalent to $(1 - \mathcal{B}_0)X = \mathcal{T}_0 X - \mathcal{B}_0 X$. Its left fixed functional is $\ell_F = \ell \circ (1 - \mathcal{B}_0)$, since $\ell_F(\mathcal{F}_0 X) = \ell(\mathcal{K}_0 X) = \ell(X) - \ell(\mathcal{B}_0 X)$. Define

$$s = \ell_F(R) = \ell(\mathcal{K}_0 R). \tag{12}$$

Because $\mathcal{K}_0 X = \mathbb{E}[(A_{u,0} - \bar{A}_0)X(A_{u,0} - \bar{A}_0)^\mathsf{T}]$, the matrix $\mathcal{K}_0 R$ is positive and $s = 1 - \ell(\mathcal{B}_0 R)$ lies between 0 and 1; for the same reason $\ell(\mathcal{B}_0 Y) \le \ell(\mathcal{T}_0 Y) = \ell(Y)$ for every $Y \succeq 0$. If s vanished, then $0 \le \ell(\mathcal{B}_0^j \mathcal{K}_0 R) \le s = 0$ for every j, and the Neumann series $\mathcal{F}_0 = \sum_{j \ge 0} \mathcal{B}_0^j \mathcal{K}_0$ would give $\ell(\mathcal{F}_0 R) = 0$, which contradicts $\mathcal{F}_0 R = R$ and $\ell(R) = 1$. So $s > 0$. The fixed space of $\mathcal{F}_0$ is spanned by R, and because $\ell_F(R) = s \neq 0$ the eigenvalue one has no Jordan chain, so it is simple. Differentiating $(1 - \mathcal{B}_\epsilon)\mathcal{F}_\epsilon = \mathcal{T}_\epsilon - \mathcal{B}_\epsilon$ at $\epsilon = 0$ and applying it to R gives $(1 - \mathcal{B}_0)\mathcal{F}_0' R = \mathcal{T}_0' R$, hence $\ell_F(\mathcal{F}_0' R) = \ell(\mathcal{T}_0' R) = -\kappa$. First-order perturbation of the simple eigenvalue then gives the order eigenvalue

$$\mu_\epsilon = 1 - \beta\epsilon + O(\epsilon^2), \qquad \beta = \frac{\kappa}{s}. \tag{13}$$

Insert $z = e^{-\lambda\epsilon}$ into Eq. (7). The other eigenvalues of $\mathcal{F}_0$ have modulus at most one, because by (iii) and the positivity argument of Appendix A.2 the spectral radius of $\mathcal{F}_\epsilon$ is below one for $\epsilon > 0$, and by the simplicity shown above they differ from one, so only μ_ϵ gives a divergent resolvent term. Since $G_1 = (1 - \mathcal{B}_\epsilon)^{-1} V_\epsilon$ and ℓ_F is taken at $\epsilon = 0$, $\ell_F(G_1) = \ell(V_\epsilon) + O(\epsilon^3) = \epsilon^2 \ell(V_*) + O(\epsilon^3)$, and

$$\begin{aligned} \sum_{k \ge 1} e^{-\lambda\epsilon k} G_k &= \frac{\epsilon\, \ell(V_*)}{s(\lambda + \beta)}\, R + O(\epsilon^2) \\ &= \epsilon\, C_*\, \frac{\beta}{\lambda + \beta} + O(\epsilon^2). \end{aligned} \tag{14}$$

Divided by Σ_ϵ from step 2 this is the Laplace transform of $\mathrm{Exp}(\beta)$, for the share of observed variance and, under the rank condition of step 3, for the ideal memory. The second identity in Eq. (7) gives $\mathbb{E}D_\epsilon = 1/(\beta\epsilon) + O(1) = s/(\kappa\epsilon) + O(1)$: s is the mixing fraction of the channel and $1/(\kappa\epsilon)$ its memory time. For the reset model, $\mathcal{T}_p = (1 - p)^2 \mathcal{T}_0$ gives $\kappa = 2$, $R = 1$ and $\ell = \mathrm{Tr}\,/n$, so $s = 1 - \|A\|_F^2/n$ and $\beta = 2/s$ as in Appendix A.4.

A.8 Numerical check

Figure 5 compares the law with exact moments. For each dimension $d = 2, 3, 4$ we draw two families of three Haar unitaries applied with probabilities $(0.2, 0.3, 0.5)$, each with two reset states (a basis state and a random pure state) and two readouts (all coordinates or one random direction). Up to $K = \lceil 6/(\beta p) \rceil$ we evaluate the remaining share $\mathrm{Tr}[\Sigma_y^+ M \mathcal{F}^K(\Sigma) M^\mathsf{T}]/q$. Over these 24 reservoirs and readouts the rates lie between 2.89 and 4.00, and the largest deviation from $e^{-\beta p K}$ falls from 0.082 at $p = 0.03$ to 0.0034 at $p = 0.001$; Fig. 5 shows one of them.

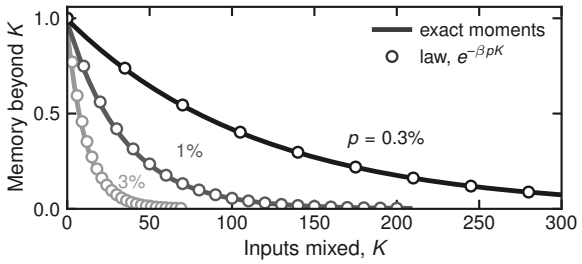


Figure 5: **The mixing law.** Share of the ideal memory carried by products of more than K inputs, for one qubit reservoir with a reset to a basis state and a scalar readout, at reset probabilities $p = 3\%$, 1% and 0.3% (light to dark). Lines: exact moments; circles: the law $e^{-\beta pK}$ with $\beta = 2/s = 2.89$.

B One qubit in closed form

During one unit of time the qubit evolves under $u\theta Z/2$ and the jump $\sqrt{\epsilon}\,|+x\rangle\langle -x|$, with signs $u = \pm 1$, and is read along Y. In the Bloch plane, $\dot{x} = -\epsilon(x-1) - u\theta y$ and $\dot{y} = u\theta x - \epsilon y/2$. For $u = +1$ write the step as

$$e^B = \begin{pmatrix} a & -b \\ b & d \end{pmatrix}, \qquad B = \begin{pmatrix} -\epsilon & -\theta \\ \theta & -\epsilon/2 \end{pmatrix}, \quad (15)$$

with offset $(f, g) = \int_0^1 e^{tB}(\epsilon, 0)\, dt$; the opposite sign reverses b and g. The stationary mean is $(m, 0)$ with $m = f/(1-a)$, and each input writes $u\,(0, v)$ with $v = bm + g$, along the measured axis.

The mean step is diagonal, while adding an input interchanges the two axes, so Y carries only odd orders. By Eq. (5) the variance passes from y to x with factor $b^2/(1-a^2)$ and back with $b^2/(1-d^2)$. With χ their product,

$$C_{2j+1} = (1-\chi)\chi^j, \qquad \mathbb{E}D = \frac{1+\chi}{1-\chi},$$
$$\operatorname{Var} Y = \frac{v^2}{(1-d^2)(1-\chi)}, \quad (16)$$

and the share of single past inputs is $d^2(1-\chi)$. At fixed θ with $\sin\theta \neq 0$, $1 - \chi = 3\epsilon/\sin^2\theta + O(\epsilon^2)$, so $\epsilon D \to \operatorname{Exp}(3/(2\sin^2\theta))$ and the share of single inputs vanishes even for exact expectation values. If instead $\theta^2/\epsilon \to \nu > 0$, then $a = 1 - (1+\nu/2)\epsilon$, $d = 1 - (1+\nu)\epsilon/2$ to first order, $b \sim \theta$, $f \sim \epsilon$ and

$$\chi \to \frac{\nu^2}{(\nu+2)(\nu+1)}, \quad \operatorname{Var} Y \to \frac{4\nu}{(\nu+2)(3\nu+2)}. \quad (17)$$

The mean order stays finite, the share of single past inputs tends to $1-\chi$, and since the recall beyond delay τ is the fraction $d^{2\tau}$, the range obeys $\epsilon\tau_{90} \to \ln 10/(1+\nu)$. The variance peaks at $\nu = 2/\sqrt{3}$. For $\theta^2 \gg \epsilon$ it vanishes as $4\epsilon/(3\theta^2)$ and for $\theta^2 \ll \epsilon$ as θ^2/ϵ, so among power laws for the angle only $\theta \propto \sqrt{\epsilon}$ keeps a finite signal. Density-matrix propagation reproduces these expressions to relative error below 10^{-12} at 25 parameter pairs.

Table 2: Strong-loss limits for X_1 in the four three-spin reservoirs: rate of the order distribution at $\eta = 1$ (certified); share of single past inputs and range at $\eta = 8/\sqrt{\gamma}$, evaluated at $\gamma = 2^{16}$.

	β_{spin}	single-input share	range/γ
H1	78.6	0.81	0.50
H2	91.7	0.29	0.18
H3	61.4	0.08	0.24
H4	132.5	0.14	0.50

C The three-spin reservoir at strong loss

Idea. At strong loss one step of the full master equation is, up to corrections of order $\epsilon = 1/\gamma$, a rotation of the dark subspace. With a fixed amplitude the two signs rotate differently; a single slow covariance builds up, and the physical-channel version of the mixing law applies. With a scaled amplitude both signs give nearly the same rotation; the signal is written on the scale of the leakage and accumulates in six slow directions, which we follow explicitly. Both arguments end in finitely many strict inequalities, which we enclose in interval arithmetic.

C.1 What is proven

For the four Hamiltonians of Table 2 and every Hamiltonian whose three couplings lie within 1/32 of theirs: with $\eta = 1$, the order of the X_1 memory divided by γ becomes exponentially distributed with a rate β_{spin}, and the variance of X_1 falls as $1/\gamma$. For the same four Hamiltonians and couplings within 1/512: with $\eta = 8/\sqrt{\gamma}$, the recall of single past inputs from X_1 tends to a positive constant, and its range divided by γ tends to a positive constant.

C.2 The slow part of a step

Let $\mathcal{D}$ be the dissipator at unit loss. Its dark space is spanned by the vacuum and the single-excitation states with coefficients $(0,1,-1)/\sqrt{2}$ and $(2,1,1)/\sqrt{6}$; the jump is injective on two and three excitations, so the kernel of $\mathcal{D}$ consists of the nine operators on the dark qutrit and all other eigenvalues have negative real part. With R embedding a Hermitian dark basis, $R^\flat = R^\mathsf{T}(\mathcal{D} + RR^\mathsf{T})^{-1}$, $\mathcal{P} = RR^\flat$ and $\mathcal{S} = (\mathcal{D}+\mathcal{P})^{-1} - \mathcal{P}$, perturbing the zero block of $\mathcal{D} + \epsilon\mathcal{H}_u$ gives

$$e^{\Delta t(\mathcal{D}/\epsilon + \mathcal{H}_u)} = RE_uR^\flat + \epsilon Z_u + O(\epsilon^2), \qquad (18)$$

with $E_u = e^{\Delta t\, R^\flat \mathcal{H}_u R}$, where Z_u contains the two boundary terms $-\mathcal{S}\mathcal{H}_u RE_uR^\flat$ and $-RE_uR^\flat\mathcal{H}_u\mathcal{S}$ and the leak $-\Delta t\, R^\flat\mathcal{H}_u\mathcal{S}\mathcal{H}_u R$ inside the exponential.

C.3 Fixed amplitude

At $\eta = 1$ the dark rotations O_u of the two signs differ. Two finite tests show that their mean is mixing and that the identity is the only fixed dark covariance: $\|\bar{O}^{4096}\|_F < 0.013$, and $1 - T_D + vv^\mathsf{T}$ is invertible, where $T_D X = \mathbb{E}[O_u X O_u^\mathsf{T}]$ and $v = 1/\sqrt{8}$. The first-order terms of Eq. (18) give the writing strength w, the decay rate κ of the slow covariance and the mixing fraction $s_0 = 1 - \|\bar{O}\|_F^2/8$, all certified positive. The physical-channel version of the mixing law then gives $\beta_{\text{spin}} = \kappa/s_0$, and the variance of X_1 is $\frac{4}{3}(w/\kappa)\epsilon + O(\epsilon^2)$ because X_1 has overlap $1/6$ with the slow covariance. Throughout the boxes of half-width $1/32$, a coupling change r moves each dark rotation by at most $3r$, and ball arithmetic keeps w, κ and s_0 above 0.0001, 3.53 and 0.052.

C.4 Scaled amplitude

At $\eta = g\sqrt{\epsilon}$, $g = 8$, both signs give the same leading rotation E. A block exponential of the static, drive and leak generators gives the mean step $E + \epsilon E_1$ and the fluctuating part $u\sqrt{\epsilon}D_1$. The six coherence phases of the static dark Hamiltonian are separated from one another and from one, so the unit-eigenvalue space of E contains only populations, on which E_1 generates a population dynamics with a unique stationary state m_*. The inputs therefore write $\pm\sqrt{\epsilon}\, v$ with $v = D_1 m_*$, which lies in the coherence planes. The fixed covariances of the rotation form a six-dimensional space: three population covariances and one identity per coherence plane. On it the first-order generators

$$\begin{aligned} K_B X &= \mathcal{P}_6(E_1' X O^\mathsf{T} + O X E_1'^\mathsf{T}), \\ K_T X &= K_B X + \mathcal{P}_6(D_1' X D_1'^\mathsf{T}) \end{aligned} \qquad (19)$$

are certified stable, and the summed linear response and the full covariance converge to $W_* = -K_B^{-1}\mathcal{P}_6(vv^\mathsf{T})$ and $\Sigma_* = -K_T^{-1}\mathcal{P}_6(vv^\mathsf{T})$. The recall of single inputs from X_1 tends to $a^\mathsf{T}W_*a/a^\mathsf{T}\Sigma_*a$, and after t/ϵ steps the remaining fraction is $a^\mathsf{T}e^{tK_B}W_*a/a^\mathsf{T}W_*a$, which crosses 0.1 once; this crossing is the range divided by γ. Interval tests bracket it to 4×10^{-6}. The certificates show $a^\mathsf{T}W_*a > 0$, and since $\Sigma_* \succeq W_*$ the limiting share is positive; the archived certificates state this numerator bound through the recall of a readout with 1000 shots, which has the same numerator. On the boxes of half-width $1/512$, Sturm counts and Weyl's bound keep the six phases separated and $-(K + K^\mathsf{T})$ positive definite for both generators, so the limits stay positive.

The single-input shares in Table 2 range from 0.08 to 0.81 because the limit compares two rates of the same order in ϵ: K_B, at which the mean step lets a stored signal decay, and the extra term of K_T, at which the random signs of later inputs turn it into products. Their ratio depends on the couplings through the dark energies and the matrix elements of X_1 between dark states, so the mean order in the limit runs from 1.4 for H1 to 5.9 for H3. The boxes are 16 times narrower than for the fixed amplitude because this argument follows the dark energy eigenstates themselves: a coupling change r moves each eigenprojector by up to $3r/(\delta - 3r)$, where δ is the smallest gap between dark energies, whereas the fixed-amplitude argument needs only the rotations, which move by at most $3r$.

C.5 Erasing coherence

Returning bright population to the dark subspace after every step leaves the leading terms of the scaled step unchanged, so it keeps the same limits. Dephasing in the dark energy basis removes the coherence planes into which v writes. Restricted to populations, the two signs then differ only at order $\epsilon^{3/2}$, because the order-$\sqrt{\epsilon}$ response of a diagonal state is off-diagonal, while the populations still follow $1 + \epsilon L$. A new input therefore writes a

vector of size $\epsilon^{3/2}$, the slow dynamics accumulate its square over $1/\epsilon$ steps, and the stored signal is $O(\epsilon^2)$.

C.6 Certificates

The attached archive `spin_certificates.zip` contains the generators, the exact couplings and the interval programs. At 192-bit precision they reproduce the archived enclosures; at 256 bits they check every strict inequality above. An independent implementation (see Author contributions) that builds the full finite-rate maps and extrapolates to infinite loss agrees with the certified rates to 2×10^{-7} and with the limiting recalls to 4×10^{-9}.

D Results of the first version

Table 3 repeats the benchmark comparison of the first version [31] at $\gamma = 1$, where the same jump families have the same total coupling. The shared channel had the largest memory and the smallest NARMA-10 error but scored below local loss on parity and on closed-loop Mackey–Glass forecasting, two tasks that need products of past inputs; at this operating point it also led local loss in memory and NARMA-10 error from four to eight spins. Figure 6 repeats its finite-sampling test: when the 45 features are estimated from a finite number of preparations per step, the advantage of the shared channel over local loss becomes significant only above about 2×10^5 preparations per step if all spins are read in three grouped settings, and only above about 3×10^6 if every feature is measured separately.

Table 3: Benchmark scores of the first version [31] at $\gamma = 1$, $h = \Delta t = 1/2$, five spins and 45 features, for parity read at 15 times within each step: mean over 32, 32, 16 and 64 reservoirs, with the standard error in the last digits in parentheses. Memory sums C_τ over delays 1 to 20 and parity sums a capacity over windows of one to seven inputs (larger is better); NARMA-10 and Mackey–Glass forecasting (MG) over 150 closed-loop steps are normalized errors (smaller is better). The rows are those of the six jump families studied here; bold marks the best of them.

family	memory	NARMA-10	parity	MG
local	8.63(7)	0.315(8)	4.82(4)	**0.037(3)**
shared	**12.21(10)**	**0.229(6)**	3.78(3)	0.097(9)
pairwise	9.20(8)	0.289(7)	4.75(4)	0.039(4)
gain/loss	8.31(6)	0.337(8)	4.85(4)	0.046(4)
exchange	7.41(5)	0.441(9)	**5.08(5)**	0.052(4)
dephasing	0	1.020(6)	0	0.0599(4)

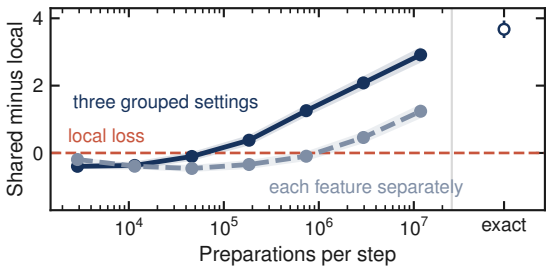


Figure 6: **Finite sampling in the first version.** Memory of one shared channel minus that of local loss at $\gamma = 1$ when every feature is estimated from a finite number of preparations per step, with all spins read along X, Y or Z in three grouped settings (solid) or each feature measured separately (dashed). Lines and bands show the mean and 95% interval over 20 reservoirs; the open circle is exact. Data from Ref. [31].

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